Brac University

Department of Electrical & Electronic Engineering

Fall2025

Course Number : EEE282

Section : **01**

Group No : **04**

Lab Report

Experiment no: 04

Name of the experiment:

Numerical Integration Techniques

Name:Souvik Barman Ratul ID:24121205

All Group members:

Sl.	***ID***	***Name***
1.	24121205	Souvik Barman Ratul
2.	24121308	Alif Tamjid
3.	23221082	Sabrina Maliha
4.	24121019	Monazzi Siddique

<u>Objective:</u>

The objective of this experiment is to learn and apply different numerical integration techniques for estimating definite integrals. We mainly focus on the Trapezoidal Rule, Simpson's 1/3 Rule, and Simpson's 3/8 Rule. Using these methods, we compute integrals of mathematical functions. This experiment helps us understand how numerical methods work .

<u>Theoretical Background:</u>

Numerical integration is used when a function is hard to integrate analytically or when only discrete data points are available. Instead of finding an exact integral, we approximate the area under a curve using simpler geometric shapes.

1.Trapezoidal Rule:

The trapezoidal rule divides the total area into trapezoids. Each pair of points forms one trapezoid and the total area is the sum of all trapezoids.

$$\int_{x_1}^{x_1+nh} f(x)dx = \frac{h}{2}(f_1 + 2f_2 + 2f_3 + \ldots\ldots 2f_n + f_{n+1})$$

2. Simpson's 1/3 Rule

Simpson's 1/3 rule approximates the curve using parabolas instead of straight lines. It requires an even number of intervals.

$$\int_{x_1}^{x_1+nh} f(x)dx = \sum_{i=1,3,5,\ldots,n-1} \frac{h}{3}(f_i + 4f_{i+1} + f_{i+2})$$

$$= \frac{h}{3}(f_1 + 4f_2 + 2f_3 + 4f_4 + 2f_5 + 4f_6 + \ \ldots\ 4f_n + f_{n+1})$$

3. Simpson's 3/8 Rule

Simpson's 3/8 rule uses cubic interpolation and needs the number of intervals to be a multiple of 3.

$$\int_{x_1}^{x_1+nh} f(x)dx = \sum_{i=1,4,7,\ldots,n-2} \frac{h}{3}(f_i + 3f_{i+1} + 3f_{i+2} + f_{i+3})$$

$$= \frac{3h}{8}(f_1 + 3f_2 + 3f_3 + 2f_4 + 3f_5 + 3f_6 + 2f_7 + \ldots + 2f_{n-2} + 3f_{n-1} + 3f_n + f_{n+1})$$

Problem Statement:

Problem-1: Write the MATLAB functions implementing Trapezoidal, Simpson's 1/3 & Simpson's 3/8 Rule to determine the integral result of any function numerically

Answer:

Code:

```
clear all;
close all;
clc;

f = @(x)exp(x); %function call
xl = 2.1; %lower limit
xh = 6.7; %higher limit

n = 12; % total boxes that fits for all rules
h = (xh-xl)./n; % difference between one point to another point

x = xl:h:xh; %increment
y = f(x); %showing the array of y for the increment x's

%Question No.1- For Trapizoide Rule

a = (h./2).*(y(1)+y(end)+2.*sum(y(2:end-1))); %Trapizoide Rule Formula

F = integral(f, xl, xh); %analytical value
error = ((a-F)/F).*100; %tolerance between the analytical and numberical value

fprintf('\nThe analytical value of this function is %.4f', F)
fprintf('\n\n\nFor Trapizoide Rule')
fprintf('\nThe area of this integer by using Trapizoide rule is %.4f', a)
fprintf('\nThe error of this analytical and numerical value in this rule is (in percentage)%.4f', error)

%Question No.1- For Simpson 1/3rd Rule

b = (h./3).*(y(1)+y(end)+4.*sum(y(2:2:end-1))+2.*sum(y(3:2:end-2))); %Simpson 1/3rd rule formula
```

```
error_1 = ((b-F)/F).*100; %tolerance between the analytical and numerical value

fprintf('\n\n\nFor Simpson 1/3rd Rule')
fprintf('\nThe area of this integer by using Simpson 1/3rd rule is %.4f', b)
fprintf('\nThe error of this analytical and numerical value in this rule is (in percentage)%.4f', error_1)
```

%Question No.1-For Simpson 3/8 Rule

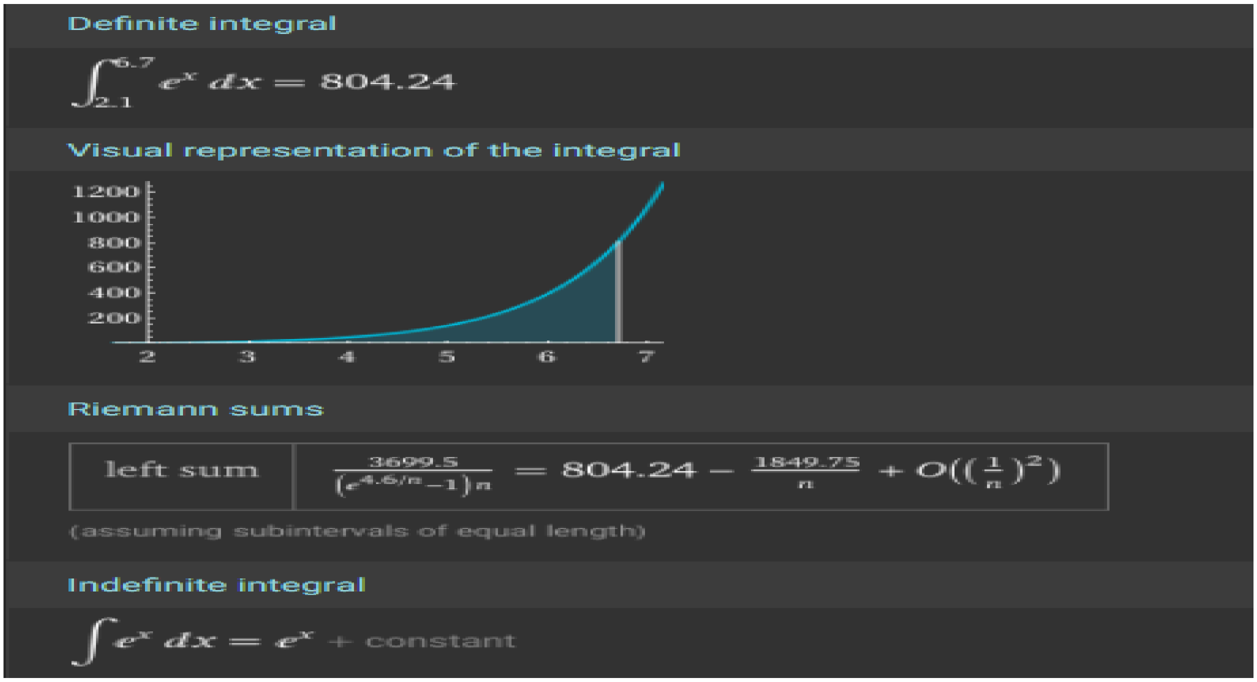
```
c = 0;
for i = 2:n
   if rem(i-1,3)==0
      c = c + 2.*y(i);
   else
      c = c + 3.*y(i);
   end
end

c = (c + y(1)+y(end)).*(3.*h/8); %Simpson 3/8 rule formula
error_2 = ((c-F)/F).*100; %tolerance between the analytical and numerical value

fprintf('\n\n\nFor Simpson 3/8 Rule')
fprintf('\nThe area of this integer by using Simpson 3/8 rule is %.4f', c)
fprintf('\nThe error of this analytical and numerical value in this rule is (in percentage)%.4f', error_2)
```

Output:

```
Command Window

The analytical value of this function is 804.2397


For Trapizoide Rule
The area of this integer by using Trapizoide rule is 814.0638
The error of this analytical and numerical value in this rule is (in percentage)1.2215


For Simpson 1/3rd Rule
The area of this integer by using Simpson 1/3rd rule is 804.3345
The error of this analytical and numerical value in this rule is (in percentage)0.0118


For Simpson 3/8 Rule
The area of this integer by using Simpson 3/8 rule is 804.4494
fx The error of this analytical and numerical value in this rule is (in percentage)0.0261>>
```

Explanation:

The code starts by clearing all variables, closing figures and clearing the windows command. After this, we define an anonymous function as where xl = 2.1 which is a lower limit and xh = 6.7 which is an upper limit. The number of intervals is defined as n=12 and the size is calculated as (xh-xl)/n which is basically h. Furthermore, we generate an array of x from xl to xh with the increment of h. The y = f(x) are computed for each x. Then we implement the Trapezoid, Simpson 1/3rd and Simpson 3/6 rule by providing the formula. Moreover, the exact integral is also calculated in this code so that we can find the accurate value.



Lastly, percentage error between Trapezoidal, Simpson 1/3rd, Simson ⅜ and exact integral is computed and printed so that we can see which method is giving more accurate value.

Discussion:-

The code demonstrates numerical integration and error analysis for different methods. It is useful for understanding how each method approximates the area under a curve. In Simpson 1/3rd rule, Odd and even interior points are summed separately in the formula. In Simpson ⅜ rule, a loop multiplies every third interior point by 2 and the others by 3. In this problem, we implemented all numerical integration methods on a known function. Since we could compute the exact result, it was easy to compare the errors. Simpson’s 1/3 and 3/8 rules gave much smaller errors than the Trapezoidal rule because they are higher-order methods. This verifies that Simpson methods generally provide better accuracy for smooth functions.

Problem-2: Determine the integral of the given function using:

$$\int_{0.5}^{2} \frac{1 + 5\sin(x^2)}{(x+1)^{\frac{3}{2}}}\, dx$$

i. Trapezoidal Rule

ii. Simpson's 1/3 Rule

iii. Simpson's 3/8 Rule

Find the Analytical result of the integral as well. Calculate the Relative Percentage Error for each of the cases.

[You may use calculator to get the analytical value of the integral]

<u>Answer</u>:

<u>Code:</u>

```
clear all;
close all;
clc;

f = @(x)((1+5.*sin(x.^2))./(x+1).^(3/2)); %function call
xl = 0.5; %lower limit
xh = 2; %higher limit

n = 12; % total boxes that fits for all rules
h = (xh-xl)./n; % difference between one point to another point

x = xl:h:xh; %increment
y = f(x); %showing the array of y for the increment x's

%Question No.2(i)- For Trapizoide Rule

a = (h./2).*(y(1)+y(end)+2.*sum(y(2:end-1))); %Trapizoide Rule Formula

F = integral(f, xl, xh); %analytical value
error = ((F-a)/F).*100; %tolerance between the analytical and numerical value

fprintf('\nThe analytical value of this function is %.4f', F)
```

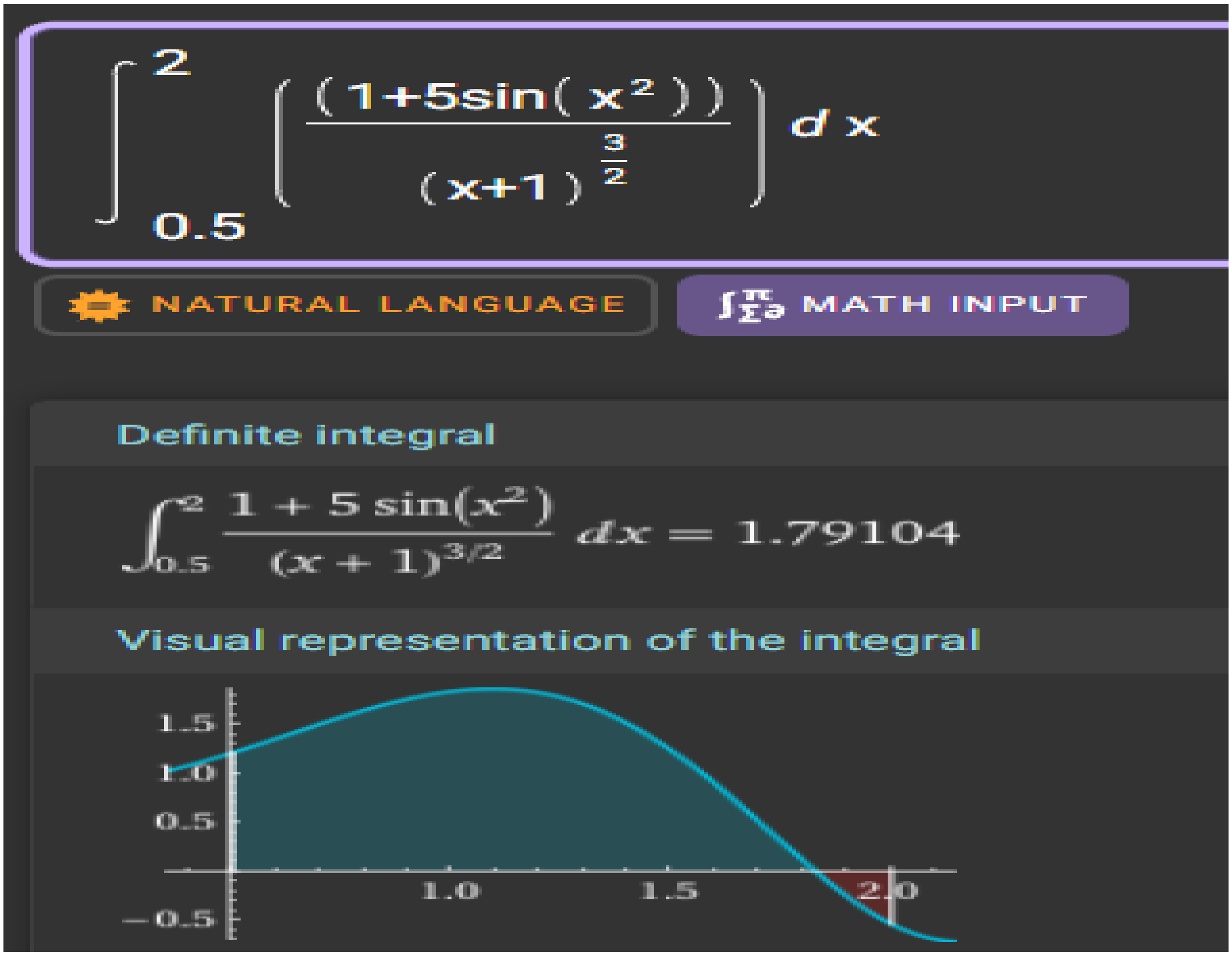
```
fprintf('\n\n\nFor Trapizoide Rule')
fprintf('\nThe area of this integer by using Trapizoide rule is %.4f', a)
fprintf('\nThe error of this analytical and numerical value in this rule is (in percentage)%.4f', error)

%Question No.2(ii)- For Simpson 1/3rd Rule

b = (h./3).*(y(1)+y(end)+4.*sum(y(2:2:end-1))+2.*sum(y(3:2:end-2))); %Simpson 1/3rd rule formula
error_1 = ((b-F)/F).*100; %tolerance between the analytical and numerical value

fprintf('\n\n\nFor Simpson 1/3rd Rule')
fprintf('\nThe area of this integer by using Simpson 1/3rd rule is %.4f', b)
fprintf('\nThe error of this analytical and numerical value in this rule is (in percentage)%.4f', error_1)

%Question No.2(iii)-For Simpson 3/8 Rule

c = 0;
for i = 2:n
   if rem(i-1,3)==0
      c = c + 2.*y(i);
   else
      c = c + 3.*y(i);
   end
end

c = (c + y(1)+y(end)).*(3.*h/8); %Simpson 3/8 rule formula
error_2 = ((c-F)/F).*100; %tolerance between the analytical and numerical value

fprintf('\n\n\nFor Simpson 3/8 Rule')
fprintf('\nThe area of this integer by using Simpson 3/8 rule is %.4f', c)
fprintf('\nThe error of this analytical and numerical value in this rule is (in percentage)%.4f', error_2)
```

Output:

```
Command Window

The analytical value of this function is 1.7910


For Trapizoide Rule
The area of this integer by using Trapizoide rule is 1.7862
The error of this analytical and numerical value in this rule is (in percentage)0.2676


For Simpson 1/3rd Rule
The area of this integer by using Simpson 1/3rd rule is 1.7911
The error of this analytical and numerical value in this rule is (in percentage)0.0040


For Simpson 3/8 Rule
The area of this integer by using Simpson 3/8 rule is 1.7912
fx The error of this analytical and numerical value in this rule is (in percentage)0.0094>>
```

Explanation:

The code starts by clearing all variables, closing figures and clearing the windows command. After this, we define an anonymous function as where xl = 2.1 which is a lower limit and xh = 6.7 which is an upper limit. The number of intervals is defined as n=12 and the size is calculated as (xh-xl)/n which is basically h. Furthermore, we generate an array of x from xl to xh with the increment of h. The y = f(x) are computed for each x. Then we implement the Trapezoid, Simpson 1/3rd and Simpson 3/6 rule by providing the formula. Moreover, the exact integral is also calculated in this code so that we can find the accurate value.



Lastly, percentage error between Trapezoidal, Simpson 1/3rd, Simson ⅜ and exact integral is computed and printed so that we can see which method is giving more accurate value.

Discussion:-

The code demonstrates numerical integration and error analysis for different methods. It is useful for understanding how each method approximates the area under a curve. In Simpson 1/3rd rule, Odd and even interior points are summed separately in the formula. In Simpson ⅜ rule, a loop multiplies every third interior point by 2 and the others by 3. In this problem, we implemented all numerical integration methods on a known function. Since we could compute the exact result, it was easy to compare the errors. Simpson's 1/3 and 3/8 rules gave much smaller errors than the Trapezoidal rule because they are higher-order methods. This verifies that Simpson methods generally provide better accuracy for smooth functions.

Problem-3: The following data are provided for the velocity, v of a moving object as a function of time, t:

t (sec.)	0	1	2	3	4
v (m/s)	0	2.15	3.26	4.18	4.82

Determine the displacements of the object for the given period of time with best possible accuracy. Which numerical approach would you choose?

[Just to recall: s = [v dt]

<u>Answer</u>:

Here,

Simpson's 1/3 Rule is the best method for determining displacement. Because data is evenly spaced and number of intervals = 4 (even, perfect for 1/3 rule) and it gives us value with small error.

<u>Code</u>:

```
clear all;

close all;

clc;

t = [0 1 2 3 4];                %% time (sec)

v = [0 2.15 3.26 4.18 4.82];    %% velocity (m/s)

xi = t(1);      %% lower limit

xf = t(end);    %% upper limit

n  = 4;     %% number of intervals

h  = (xf - xi)/n; %% step size


a = h/3*( v(1) + v(end) + 4*sum(v(2:2:end-1)) + 2*sum(v(3:2:end-2)));

fprintf("Displacement = %f m\n", a);
```

Output:

```
Command Window
  Displacement = 12.220000 m
>>
```

Explanation:

The code calculates the displacement of an object from its velocity data using Simpson's 1/3 rule. The time data t and velocity data v are given at discrete points. The total time interval is divided into n = 4 equal parts with step size h. Simpson's 1/3 formula is applied to approximate the area under the velocity-time curve, which represents the displacement. Finally, the result is printed in meters.

Discussion:

In this problem, we were given velocity values at five evenly spaced time points. Since the data points are equally spaced and the number of intervals is 4 (which is even), Simpson's 1/3 Rule is the most suitable numerical method. This rule usually gives better accuracy than the Trapezoidal Rule because it uses parabolic curves to estimate the area, not straight lines. After applying Simpson's 1/3 Rule, we were able to find the displacement by calculating the area under the velocity-time graph. Because the velocity changes smoothly, Simpson's 1/3 Rule works very well here and gives a reliable result. Overall, this method was the best choice for getting the most accurate displacement from the given data.

Problem-4: Consider the data-points once again:

t (sec.)	0	1	2	3	4	5	6
v (m/s)	0	2.15	3.26	4.18	4.82	5.61	4.77

Determine the displacements again. Which numerical approach would you choose now to have better accuracy?

Answer:

Here,

Simpson's 1/3 Rule is the best method for determining displacement. Because data is evenly spaced and number of intervals = 6 (even, perfect for 1/3 rule) and it gives us value with small error.

Code:

```
clear all;

close all;

clc;

t = [0 1 2 3 4 5 6];  %% time (sec)

v = [0 2.15 3.26 4.18 4.82 5.61 4.77];  %% velocity (m/s)

xi = t(1);          %% lower limit

xf = t(end);        %% upper limit

n  = 6; %% number of intervals

h  = (xf - xi) / n; %% step size



a = (h/3) * (v(1)+ v(end)+ 4*sum(v(2:2:end-1))+ 2*sum(v(3:2:end-2)));


fprintf("Displacement =%f m\n", a);
```

Output:

Command Window

```
Displacement = 22.896667 m
>>
```

Explanation:

The code calculates the displacement of an object using velocity data at discrete time points with Simpson's 1/3 rule. The time t and velocity v arrays are given, and the total interval is divided into n = 6 equal parts. The formula combines the first and last velocity values with weighted sums of the intermediate points (4 for odd indices, 2 for even indices) to approximate the area under the velocity-time curve. This area represents the displacement, which is then printed in meters.

Discussion:

For this problem, the velocity data included seven time points, which created 6 intervals. Since 6 is an even number, Simpson's 1/3 Rule can again be used. Just like in Problem 3, the data is evenly spaced, which makes the method appropriate. Simpson's 1/3 Rule gives better accuracy than the Trapezoidal Rule because it fits a parabola through the data instead of straight line segments. The velocity values also change smoothly here so the method provides a very good estimate of the displacement. By applying the rule we calculated the total area under the velocity-time graph, which represents the displacement. In summary due to the even number of intervals and smooth data, Simpson's 1/3 Rule is the best method for this problem as well.